Customer Retention & Revenue Growth Strategy

Telecom Industry | Executive Analysis Report

An end-to-end descriptive analysis of 7,043 fictional telecom customer records using Excel, MySQL, and Power BI.

Total customers	Churn rate	Monthly revenue	Revenue at risk
7,043	**26.54%**	**$456,116.60**	**$139,130.85**

Business question

Where is churn concentrated, how much monthly recurring revenue is associated with churned customers, and which retention actions should management test first?

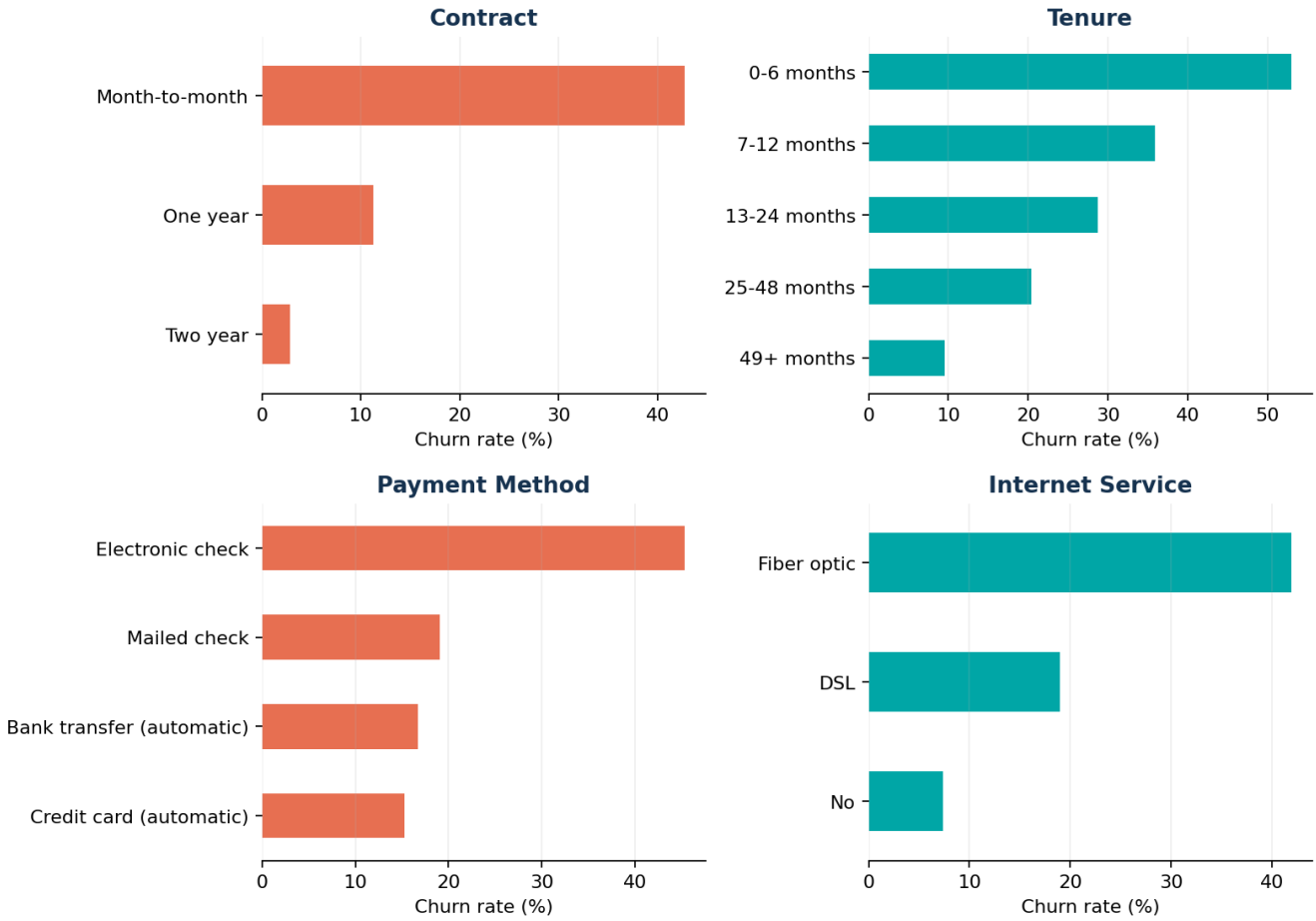
Key findings

Theme	Evidence
Contract	Month-to-month churn was 42.71%, compared with 11.27% for one-year and 2.83% for two-year contracts.
Tenure	Customers in months 0-6 churned at 52.94%; customers with 49+ months churned at 9.51%.
Payments	Electronic-check users had the highest payment-method churn rate at 45.29%.
Service	Fibre-optic customers churned at 41.89%, while customers without tech support and online security churned at 41.64% and 41.77%.

Interpretation

The patterns identify useful targeting hypotheses, but they do not prove causation. “Revenue at risk” is the sum of monthly charges for customers recorded as churned; it is not a realised-loss or forecast measure.

Segment evidence



Recommended retention experiments

Action	Test
1. First-90-day onboarding	Provide setup guidance and proactive support to short-tenure, month-to-month fibre customers.
2. Contract migration	Test modest incentives for suitable month-to-month customers to move to annual plans.
3. Support and security bundles	Offer targeted trials to customers without tech support or online security.
4. Payment journey	Investigate electronic-check friction and test autopay migration messaging.
5. Measurement	Track incremental churn reduction, offer uptake, cost per retained customer, and protected recurring revenue by cohort.

Data quality and limitations

The raw 21-column file was preserved. Eleven blank TotalCharges values belonged to zero-tenure customers and were converted to 0. All 7,043 customer IDs are unique, and the cleaned data contains no missing values. The dataset is fictional, contains no event dates, and supports descriptive segment analysis rather than time-series, causal, or predictive claims.

Source: IBM Telco Customer Churn sample dataset.